



Vinitha Pukazh Bagya R <pukazhvinitha@gmail.com>

Your application as Senior Backend Web Developer - next steps

8 messages

Nikolaus Kirstein <nk@principiamentis.com>
To: "pukazhvinitha@gmail.com" <pukazhvinitha@gmail.com>
Cc: Andreas Bergmaier <ab@principiamentis.com>

Sat, May 16, 2026 at 1:01 AM

Hi Vinitha,

hope you're doing well. Your first interview left us with a very positive impression and we would like to continue the interview process with you. The next step would be a technical interview. However, as I think we've mentioned before, we haven't yet decided how exactly we want to do this, i.e. whether it would only be an interview or whether it would include some sort of task or challenge for you to complete. As you know, we're doing this for the first time, and both of us have other work to do (like, actually write code), so we'll need a few days to figure this out. We will get back to you with more concrete information by Thursday of next week.

Thanks and kind regards,
Nikolaus

Nikolaus Kirstein
Software Development

Principia Mentis GmbH
[Paradiesstr. 9](#), Gartenhaus, 80538 München, Deutschland
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vinitha pukazh bagya <pukazhvinitha@gmail.com>
To: Nikolaus Kirstein <nk@principiamentis.com>
Cc: Andreas Bergmaier <ab@principiamentis.com>

Sun, May 17, 2026 at 10:11 AM

Hi Nikolaus,

Thank you so much for the update — I'm really glad to hear the first interview went well, and I'm excited to continue the process!

Please take all the time you need to figure out the format. I'm happy to work with whatever you decide. I look forward to hearing from you by Thursday.

Thanks again

Warm regards,
Vinitha

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Nikolaus Kirstein <nk@principiamentis.com>
To: vinitha pukazh bagya <pukazhvinitha@gmail.com>
Cc: Andreas Bergmaier <ab@principiamentis.com>

Thu, May 21, 2026 at 11:57 PM

Hi Vinitha,

thank you for your patience and understanding! For the next step in the interview process, we would like to give you a technical challenge, which you can complete in your own time, and then do a longer interview (approx. 2 hours) in which we will discuss the challenge and also ask you some technical questions regarding JavaScript/TypeScript, NodeJS, SQL and web technologies in general.

The challenge is as follows:

Consider a tree structure where each node represents a task or an area of work in a project. Nodes hold data like start and end dates, text, tags, and numeric KPIs.

1. How would you design a relational database schema for such a tree structure? You can ignore the data fields and focus on the relationships between the nodes.
2. How do you write a recursive SQL query in the schema?
3. How would you design a schema for a system of roles and permissions on this tree structure that meets the following requirements:
 - Users are assigned in roles at nodes, and each role assignment comes with certain permissions at the node where the user is assigned. Some roles also include permissions on the descendants and/or ancestors of the node where the user is assigned.
 - The roles to be considered are: Admin, Requester, Responsible, Viewer
 - The permissions to be considered are:
 - ViewStructure: permission to view the node (i.e. see that it exists, e.g. in a graphical representation of the tree)
 - EditStructure: permission to move/delete the node and add children to it
 - ViewData: permission to view the data stored at the node
 - EditData: permission to edit the data stored at the node
 - The relationships between the roles and permissions are as follows:
 - A user assigned as Admin at node N has all permissions at N and at all descendants and ancestors of N
 - A user assigned as Requester at node N has all permissions at N and on all descendants of N, and ViewStructure and ViewData permissions on all ancestors of N
 - A user assigned as Responsible at node N has all permissions at N and on all descendants of N, and ViewStructure permission on all ancestors of N
 - A user assigned as Viewer at node N has ViewStructure and ViewData permissions at N and at all descendants of N, and ViewStructure permission on all ancestors of N

As you may have guessed, this is a simplified version of a problem we faced ourselves when building our application.

Please do not spend more than four hours on this, we don't want to abuse your time. As output, please send us an SQL file with your proposed schema definition so we can look at it before the interview. For the interview itself, it would be fantastic if you could set up a small demo database to run some actual queries against, but this is entirely optional. We're mostly interested in your thought process, the questions you encounter along the way, the assumptions, decisions and trade-offs you make, etc., so those are the things we will focus on in the interview. If you use AI, we would appreciate it if you were open about it.

We can schedule the interview for the first week of June so you have some time to complete the challenge. I will contact you again next week with possible dates and times.

In the meantime, please do not hesitate to contact us if you have any questions.

Kind regards,
Nikolaus

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Software Development

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Von: vinitha pukazh bagya <pukazhvinitha@gmail.com>
Gesendet: Sonntag, 17. Mai 2026 06:41
An: Nikolaus Kirstein <nk@principiamentis.com>
Cc: Andreas Bergmaier <ab@principiamentis.com>
Betreff: Re: Your application as Senior Backend Web Developer - next steps

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vinitha pukazh bagya <pukazhvinitha@gmail.com>
To: Nikolaus Kirstein <nk@principiamentis.com>
Cc: Andreas Bergmaier <ab@principiamentis.com>

Fri, May 22, 2026 at 5:22 AM

Hi Nikolaus,

Thank you for the details. This looks like a genuinely interesting challenge.

I'm happy to take it on and will aim to submit the SQL file well ahead of our interview. The first week of June works well for me. I'll wait to hear from you with the available dates and times.

Looking forward to it!

Best regards,
Vinitha

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Nikolaus Kirstein <nk@principiamentis.com>
To: vinitha pukazh bagya <pukazhvinitha@gmail.com>
Cc: Andreas Bergmaier <ab@principiamentis.com>

Wed, May 27, 2026 at 6:56 PM

Hi Vinitha,

hope you're doing well. The following dates and times are possible from our side for the technical interview:

- Monday, June 1, any time after 15:00 CET
- Tuesday, June 2, any time after 15:00 CET
- Wednesday, June 3, at 11:00 or after 15:00 CET

Would any of these work for you?

Best regards,
Nikolaus

Nikolaus Kirstein
Software Development

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Von: vinitha pukazh bagya <pukazhvinitha@gmail.com>

Gesendet: Freitag, 22. Mai 2026 01:52

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Nikolaus Kirstein <nk@principiamentis.com>

Wed, May 27, 2026 at 8:40 PM

To: vinitha pukazh bagya <pukazhvinitha@gmail.com>

Hi Vinitha,

quick update: the slots on Wednesday, June 3, are no longer available. Hope you can make one of the others.

Best regards,
Nikolaus

Nikolaus Kirstein
Software Development

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Von: Nikolaus Kirstein <nk@principiamentis.com>

Gesendet: Mittwoch, 27. Mai 2026 15:26

An: vinitha pukazh bagya <pukazhvinitha@gmail.com>

Cc: Andreas Bergmaier <ab@principiamentis.com>

Betreff: AW: Your application as Senior Backend Web Developer - next steps

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vinitha pukazh bagya <pukazhvinitha@gmail.com>

Wed, May 27, 2026 at 10:41 PM

To: Nikolaus Kirstein <nk@principiamentis.com>

Hi Nikolaus,

Thank you for the update. Tuesday, June 2 at 15:00 CET works perfectly for me.

Regarding the assessment, I'll be sharing my output in the coming days, I wasn't able to focus on it earlier as I was unwell.

Looking forward to the interview!

Best regards,
Vinitha

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Nikolaus Kirstein <nk@principiamentis.com>

Wed, May 27, 2026 at 11:00 PM

To: vinitha pukazh bagya <pukazhvinitha@gmail.com>, Andreas Bergmaier <ab@principiamentis.com>

Hi Vinitha,

perfect, I'll send you a Teams invite for Tuesday. I'm sorry to hear you've been unwell, hope you get better soon!

Looking forward to speaking to you again.

Best regards,
Nikolaus

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Software Development

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Von: vinitha pukazh bagya <pukazhvinitha@gmail.com>

Gesendet: Mittwoch, 27. Mai 2026 19:11

An: Nikolaus Kirstein <nk@principiamentis.com>

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